

Variation-aware proxy learning for semantic segmentation

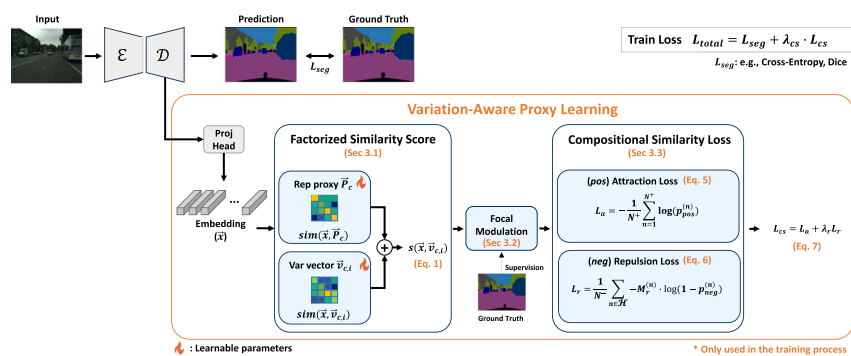
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HIGHLIGHTS

- Compositional Similarity Loss enables the learning of representative proxies and variation vectors.
- Factorized similarity score achieves enhanced class separation with proxy integration.
- Focal modulation adaptively emphasizes ambiguous regions with dynamic weighting.
- Consistent segmentation gains across diverse real-time and non-real-time models.

GRAPHICAL ABSTRACT



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ABSTRACT

In semantic segmentation, accurately modeling **intra-class variation** is essential for capturing fine-grained details and resolving ambiguity near class boundaries. While **existing proxy-based embedding methods** represent each class with a single prototype, they struggle to reflect diverse intra-class structures, especially in complex scenes. In this paper, we propose a novel representation learning framework called Variation-Aware Proxy Learning, which introduces a **representative proxy to encode shared class semantics and multiple variation vectors to capture fine-grained intra-class variations**. These components are integrated through a factorized **similarity score**, enabling more expressive and discriminative embedding structures. To further enhance learning in ambiguous regions, we introduce focal modulation and design a new Compositional Similarity Loss composed of attraction and repulsion terms that adaptively amplify the contribution of hard examples. Our method is model-agnostic and requires no additional inference-time cost. Extensive experiments across multiple segmentation benchmarks—Cityscapes, COCO-Stuff10k, iSAID and ADE20K—and diverse backbones including CNNs and Transformers, demonstrate consistent improvements in mIoU and boundary precision, particularly in challenging regions with high intra-class variability.

1. Introduction

Semantic segmentation aims to assign predefined labels to every pixel in an image, enabling a comprehensive understanding of the structure and relationships among scene components. Early methods relied on classical image processing and geometric models such as

edge detection and region-based segmentation. However, the advent of deep learning has significantly advanced the field [1,8,10]. Deep models, empowered by large-scale datasets and powerful feature extractors, have greatly improved prediction accuracy and demonstrated their effectiveness in diverse real-world applications, including autonomous driving [1,2,23], medical imaging [10], and remote sensing [24–26].

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To mitigate such ambiguity, recent studies have explored class-wise relational reasoning and feature refinement [27]. Among them, contrastive learning has shown promise in enhancing class discrimination by pulling semantically similar features together and pushing dissimilar ones apart [4,5]. However, existing contrastive approaches often rely on heuristic sampling strategies or pixel-wise embeddings, which may be insufficient to capture semantic-level structures and lead to unstable training. Moreover, applying uniform contrastive constraints across all regions fails to account for spatial heterogeneity—particularly in ambiguous boundary areas where stronger semantic guidance is needed. These limitations highlight the need for a more adaptive and context-aware contrastive framework for semantic segmentation.

To address this issue, recent works have proposed restructuring pixel features in a high-dimensional embedding space, encouraging intra-class compactness and inter-class separation [4,5]. **A common approach [6,7] involves learning a single prototype per class and comparing all pixel embeddings against it. However, this strategy struggles to represent diverse intra-class variations, making it less effective when pixel embeddings reflect non-representative or auxiliary characteristics of the class.**

The limitation of using a single proxy per class lies in its inability to capture intra-class diversity, making it difficult to distinguish between representative and auxiliary characteristics within the embedding space. To overcome this, we propose a dual-component approach that introduces a representative proxy to capture shared class semantics and a set of variation vectors to model intra-class variation. We compute a factorized similarity score by separating and integrating the similarity between embeddings and each component, enabling more precise relational modeling. Based on this score, we design an Attraction/Repulsion Loss that pulls embeddings closer to their correct class while pushing away misclassified ones. This formulation allows clearer separation of semantically similar class boundaries and reduces the influence of outliers and noise through refined cosine similarity alignment, ultimately leading to significantly improved segmentation performance.

The main contributions of this paper are summarized as follows:

- We introduce **variation vectors**, a novel modeling construct that captures intra-class variation in semantic segmentation, enabling more expressive per-class representations.
- We propose a **factorized representation** that jointly encodes shared semantics and class-specific variations, enhanced by focal modulation to better distinguish subtle features near semantic boundaries.
- Our method is designed in a **model-agnostic manner**, allowing it to be seamlessly integrated into various existing segmentation frameworks, thereby offering strong generalizability across tasks.

2. Related work

2.1. Semantic segmentation

The field of semantic segmentation has seen continuous progress through a wide range of architectural innovations in recent years [1–3,8–10,13,28,30–33,44,46–48]. Early studies focused on establishing end-to-end architectures for pixel-wise prediction. Notably, models such as FCN [1] and U-Net [10] adopted encoder–decoder structures, effectively restoring local image details and capturing fine-grained spatial structures.

Subsequent research has focused on more sophisticated architectural designs to better exploit global contextual information. For example, the DeepLab series [8,9] and PSPNet [2] integrated multi-scale feature extraction techniques—such as atrous convolution, ASPP, and pyramid pooling modules—to effectively combine local and global context, improving overall accuracy. DeepLab enhanced boundary quality and prediction consistency by capturing information at multiple scales, while PSPNet maximized the use of global context. Building on this foundation, recent studies have further strengthened global context modeling by incorporating attention mechanisms into CNN backbones [30,31,34]. For

instance, [34] inserted channel and spatial attention modules into multi-resolution CNN blocks to capture complex spatial relationships, and [35] proposed a lightweight U-Net variant with multi-attention fusion to achieve both memory efficiency and improved context understanding.

Despite these efforts, CNN-based approaches still face limitations in capturing long-range dependencies and global context effectively [12]. To overcome this, Transformer-based architectures have recently gained traction [3,13,28], leveraging self-attention mechanisms to model long-range relationships. For instance, SETR [3] directly captures global features, while SegFormer [13] combines a lightweight decoder with a Transformer backbone for a balance of efficiency and performance. TransUNet [28] fuses CNN and Transformer modules to achieve stable and robust predictions, and AFFormer [32] focuses on extremely lightweight segmentation using a prototype-based approach. In the medical domain, HiFormer [33] adopts a hierarchical Transformer design to effectively integrate global context with local details.

However, existing semantic segmentation approaches still struggle to resolve confusion between semantically similar classes [2,8,12]. In particular, pixel-wise classification often focuses on optimizing relative class probabilities rather than enforcing clear inter-class separability, which makes it difficult to distinguish closely related categories.

2.2. Rethinking proxy-based embedding

Conventional semantic segmentation methods have primarily focused on observable features to separate classes. However, such approaches often struggle to ensure clear inter-class separation, particularly when visually similar classes are involved [4,6,7,14–22,29,57]. To address this, a growing body of research has explored embedding-based approaches that explicitly enforce class separation in the embedding space. These methods can be broadly categorized into three directions: contrastive learning [4], hierarchical segmentation [14–16], and efficient proxy-based embedding techniques [6,7,22,29].

Contrastive learning approaches aim to directly optimize semantic similarity between pixel embeddings by pulling together those from the same class while pushing apart those from different classes [4,57]. For instance, Wang et al. [4] proposed a pixel-level contrastive loss that effectively improves inter-class discrimination. However, the computational cost of pairwise comparisons becomes prohibitive as image resolution increases.

Hierarchical segmentation methods have also proven effective by explicitly modeling inter-class relationships. Liang et al. [14] used a Semantic Neuron Graph to encode class-level hierarchies in the embedding space. Li et al. [15] further improved pixel-level class separation by learning hierarchical structures, and Weber et al. [16] employed hyperbolic embeddings to represent class hierarchies more effectively than Euclidean space. More recent work has additionally incorporated logical reasoning into segmentation to ensure predictions respect hierarchical and relational consistency [59].

Despite their effectiveness, contrastive learning often suffers from high computational overhead, and hierarchical methods rely on predefined class structures, limiting their general applicability. To overcome these challenges, proxy- or prototype-based embedding techniques have gained attention [6,7,22,29]. Proxy-based metric learning methods such as Proxy-NCA [6] and Proxy Anchor Loss [7] introduce class-wise representative vectors to improve computational efficiency while preserving embedding separability. In parallel, clustering-based formulations have also been explored, where pixel embeddings are iteratively assigned to learnable cluster centers rather than relying on fixed proxy initialization [58].

However, these approaches primarily focus on inter-class separation and tend to overlook intra-class variation. Recent advances have thus explored fine-grained prototype modeling to better capture within-class variability. For example, Prototypical Networks [29] enhance class-wise embedding structure by learning class-specific prototypes, and Zhou et al. [22] extend this idea to better handle intra-class variations. In

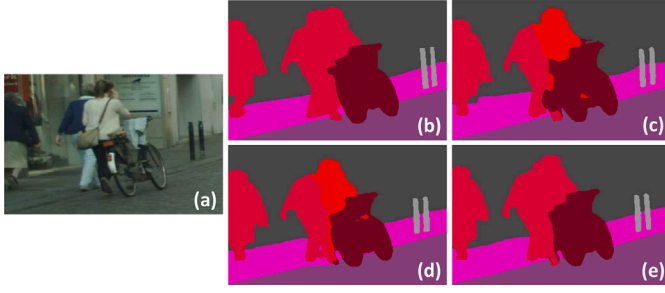


Fig. 1. (a) Input image, (b) Ground Truth, (c) Without proxy-based method, (d) With separated variation vectors (prior multi-prototype), (e) Proposed: jointly learned variation vectors. (e) shows clearer separation between semantically similar classes by capturing both shared and distinct characteristics via the representative proxy and variation vectors.

particular, Zhou et al. [22] addressed intra-class variation in semantic segmentation by introducing multiple sub-prototypes per class, allowing the model to better capture diverse class-specific characteristics. However, this approach heavily relies on prototype initialization and clustering procedures, making the optimization process more complex and sensitive to hyperparameters such as the momentum update coefficient. For instance, online updates can cause prototypes to fluctuate due to small shifts in embedding distributions at each iteration. To mitigate such instability, additional constraints like equipartition and carefully tuned hyperparameters are required. These limitations are also reflected in the results shown in Fig. 1 (“separated sub proxies”), indicating that prior methods still leave substantial room for improvement.

3. Method

To effectively capture the diverse intra-class characteristics that a single proxy cannot fully represent, this paper proposes a novel representation learning scheme for semantic segmentation based on a new loss formulation called Compositional Similarity Loss. As illustrated in Fig. 2, variation vectors are introduced to model intra-class variation for each class. Each variation vector ($\vec{v}_{c,i}$) contributes to a factorized similarity score, which accurately measures the similarity between embeddings by jointly reflecting both representative and fine-grained class-specific traits. From a mixture-modeling standpoint, each variation vector $\vec{v}_{c,i}$ acts like an individual component in a Gaussian mixture model—allocating capacity to capture local intra-class modes

that a single proxy cannot—thereby enriching the representation with multiple, distinct sub-distributions. To further enhance discrimination near semantic boundaries, we incorporate a focal modulation mechanism that strengthens learning signals in ambiguous regions. Based on these components, the proposed **Attraction Loss** encourages embeddings to remain close to both the representative proxy and corresponding variation vectors of the correct class, while the Repulsion Loss penalizes embeddings of misclassified samples when their similarity exceeds a predefined margin. Together, these mechanisms enable clearer separation between classes and more precise semantic boundaries.

As illustrated in Fig. 2, the overall training pipeline begins by feeding an input image through an encoder–decoder architecture to produce a pixel-wise prediction mask. A standard loss function, such as Cross-Entropy (CE) or Dice loss, is then computed between the predicted and ground truth (GT) masks. Simultaneously, **pixel embeddings** extracted during the forward pass are passed through a projection head. The Compositional Similarity Loss is then computed from their similarities to the representative proxy and the variation vectors and added to the total training objective. Importantly, the proposed method focuses on enhancing the structure of the embedding space only during training. At inference time, it relies solely on the base encoder–decoder network, incurring no additional computational overhead. Thus, this model-agnostic, joint learning framework effectively improves semantic segmentation by promoting intra-class compactness and inter-class separability within the embedding space Fig. 3.

3.1. Factorized similarity score

Traditional metric learning and classification-based embedding methods typically employ a single proxy or prototype (hereafter referred to as proxy) to represent each class in the embedding space. While effective at capturing a class’s central tendency, this approach falls short when classes exhibit diverse distributions or multi-modal patterns—an issue especially critical in semantic segmentation, where fine-grained boundaries and structural details must be preserved at the pixel level. In such cases, a single proxy often fails to account for intra-class variations, leading to degraded performance near class boundaries or in complex regions.

To address this limitation, we propose a factorized similarity score that more precisely models the relationship between embeddings and proxies. It combines two complementary components: **a representative proxy** capturing the core semantics of each class, and **a set of variation vectors modeling finer intra-class variations**. Unlike prior

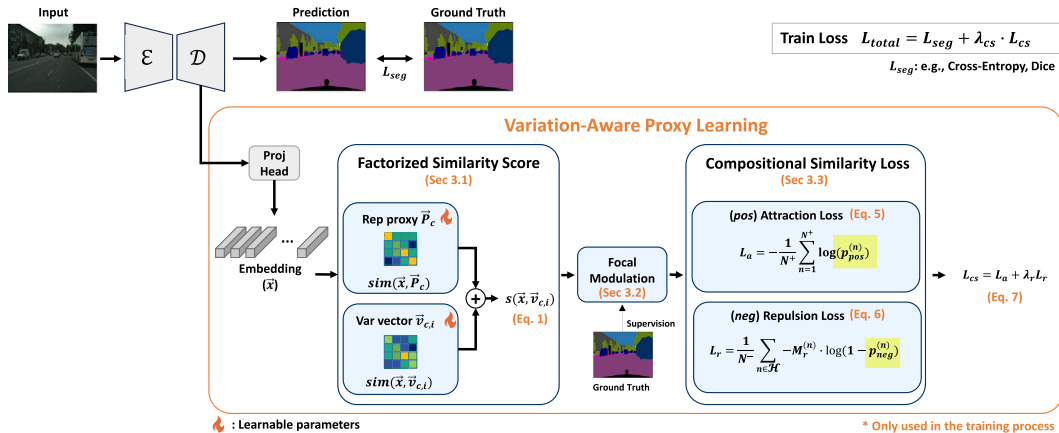


Fig. 2. Overview of the proposed training pipeline. Embeddings from the projection head compute similarities with representative proxies and variation vectors, forming a factorized similarity score. Focal modulation is applied to hard examples to compute attraction loss (L_a) and repulsion loss (L_r), which together form the Compositional Similarity Loss (L_{cs}). Here, M_r represents the focal weight for hard examples.

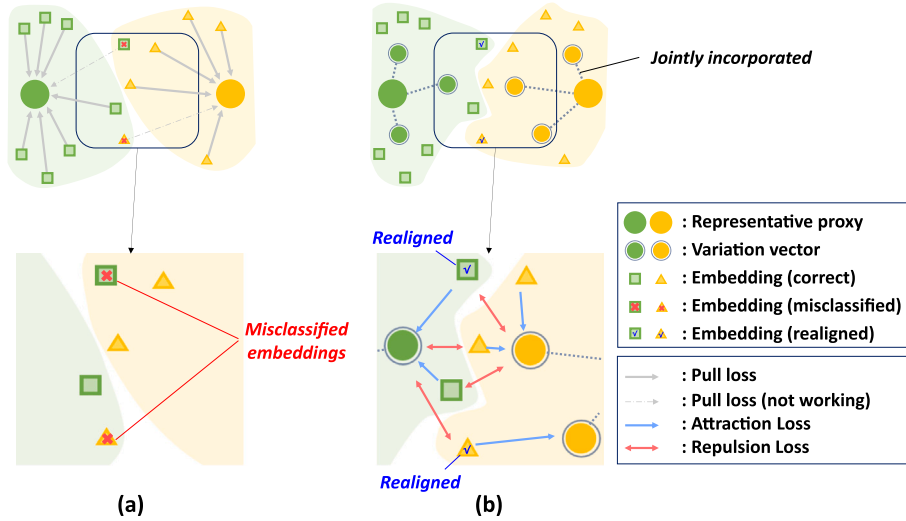


Fig. 3. (a) Single class proxy per class vs. (b) Proposed Variation-Aware Proxy with representative proxy and variation vectors. Our method captures multiple distinct characteristics per class, enhancing robustness to non-dominant features and improving separability near class boundaries. In (b), dotted lines indicate that the factorized similarity score jointly incorporates similarities to both representative proxy and variation vectors for loss computation.

multi-proxy approaches that treat proxies independently—often resulting in redundant representations or inefficient mode separation—our formulation assigns distinct roles to each component, yielding a more balanced and expressive representation. Meanwhile, all embedding vectors, representative proxies, and variation vectors are maintained as unit norm through L2 normalization. This removes distortions caused by vector magnitude and enables stable, direction-based similarity comparisons. The representative proxies and variation vectors are initialized using Xavier initialization and are subsequently updated end-to-end during training. Finally, the factorized similarity score is defined by combining the similarity between the representative proxy $\bar{\mathbf{P}}_c$ and the embedding $\bar{\mathbf{x}}$, and the similarity between the corresponding variation vector $\bar{\mathbf{v}}_{c,i}$ and $\bar{\mathbf{x}}$, as follows:

$$s(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i}) = \text{sim}(\bar{\mathbf{x}}, \bar{\mathbf{P}}_c) + \lambda_{var} \cdot \text{sim}(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i}) \quad (1)$$

Here, $c \in \{1, \dots, C\}$ denotes the class index, and $i \in \{1, \dots, K_c\}$ indexes the variation vectors of class c , with K_c their count.

The proposed method splits the similarity score into a global proxy term and a local variation term, boosting the mutual information between embeddings and their true labels while constraining intra-class variation—thereby achieving both strong class separation and coherent sub-mode alignment. Here, the hyperparameter λ_{var} is introduced to control the relative contribution of the variation vectors in the final similarity score. Here, we set $\lambda_{var} = 1.0$, and $\text{sim}(a, b)$ indicates the cosine similarity between a and b . This value is theoretically bounded above by the maximum similarity of $1 + \lambda_{var}$. Here, we set $K_c = 5$ as the default configuration.

3.2. Focal modulation

The proposed factorized similarity score captures both the common characteristics of each class via the representative proxy and the intra-class variations via the variation vectors. However, simple similarity-based scoring may lack sensitivity to **hard examples** near class boundaries. To address this, focal modulation is incorporated by adopting the focusing term $(p_i)^\gamma$ from the focal loss $FL(p_i) = -\alpha_{FL}(1 - p_i)^\gamma \log(p_i)$ [36], applied only to negative samples to emphasize hard negatives. We **omit the positive focusing term $(1 - p_i)^\gamma$ to prevent vanishing gradient issues when p_i approaches 1**. Within this framework, the probability p_i is instantiated as the variation vector probability distribution $p_{\text{sub}}(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i})$.

Specifically, $p_{\text{sub}}(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i})$ is computed by scaling the factorized similarity score $s(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i})$ —which combines similarities with the representative

proxy and its variation vectors—by τ (set to 10.0) and passing it through a softmax:

$$p_{\text{sub}}(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i}) = \frac{\exp(\tau \cdot s(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,i}))}{\sum_{j=1}^{K_c} \exp(\tau \cdot s(\bar{\mathbf{x}}, \bar{\mathbf{v}}_{c,j}))} \quad (2)$$

Pixels are indexed by n , and $y^{(n)} \in \{1, \dots, C\}$ denotes the ground-truth class of pixel n . For pixel n , let $c^* = y^{(n)}$ and $i^* = \arg \max_i s(\bar{\mathbf{x}}^{(n)}, \bar{\mathbf{v}}_{c^*,i})$. The positive index set for pixel n , denoted $\mathcal{I}_{\text{pos}}^{(n)}$, is defined as $\{(c^*, i) \mid i = 1, \dots, K_{c^*}\}$, i.e., all variation vector indices within the ground-truth class. The negative index set is $\mathcal{I}_{\text{neg}}^{(n)} = \{(c, i) \mid c \neq c^*\}$, i.e., all variation vector indices from non-ground-truth classes. The negative-variation probability for pixel n is defined as:

$$p_{\text{neg}}^{(n)} = \max_{(c,i) \in \mathcal{I}_{\text{neg}}^{(n)}} p_{\text{sub}}(\bar{\mathbf{x}}^{(n)}, \bar{\mathbf{v}}_{c,i}) \quad (3)$$

Based on $p_{\text{neg}}^{(n)}$, the per-pixel modulation factor is defined as:

$$M_r^{(n)} = (p_{\text{neg}}^{(n)})^\gamma \quad (4)$$

Here, γ is the focusing parameter that amplifies the loss contribution for hard negative examples; we set $\gamma = 2.0$ as the default value. When a negative sample yields a high probability, $M_r^{(n)}$ amplifies its loss contribution to suppress representations that resemble the target class.

By incorporating negative-only focal modulation into the factorized similarity score, the model emphasizes hard negatives and refines decision boundaries more precisely near class interfaces. Specifically, the modulation factor $M_r^{(n)}$ amplifies the loss contribution for negative variation vectors that exhibit high similarity to the target class, thereby suppressing errant variation vectors that cluster incorrectly. As a result, this targeted focusing enhances inter-class separability by pushing divergent variation vectors away from incorrect clusters, while preserving intra-class compactness via the unchanged attraction term. This design yields a more robust embedding space that captures subtle boundary uncertainties without relying on positive focusing, effectively improving segmentation performance in challenging regions.

3.3. Compositional similarity loss

The core components of the proposed scheme—**Attraction Loss** and **Repulsion Loss**—serve to finely regulate the embedding distribution for

positive and negative samples, respectively. These losses are defined based on the variation vector probability distribution $p_{\text{sub}}(\vec{x}, \vec{v}_{c,i})$ and the negative-only focal modulation factor M_r introduced in Section 3.2.

First, the Attraction Loss L_a is formulated without positive focal modulation to avoid vanishing gradients when the correct-class probability is high. Aggregating the ground-truth class into a single positive probability yields $p_{\text{pos}}^{(n)} = \max_{(c,i) \in \mathcal{I}_{\text{pos}}^{(n)}} p_{\text{sub}}(\vec{x}^{(n)}, \vec{v}_{c,i})$. The attraction term is then defined as:

$$L_a = -\frac{1}{N^+} \sum_{n=1}^{N^+} \log(p_{\text{pos}}^{(n)}) \quad (5)$$

Here, N^+ denotes the number of pixels, and $N^- = |\mathcal{H}|$ denotes the number of hard-negative pixels selected by $R^{(n)} > \tau_R$.

In contrast, the Repulsion Loss L_r penalizes high similarity to negatives by amplifying $-\log(1 - p_{\text{neg}}^{(n)})$ with the focal weight $M_r^{(n)} = (p_{\text{neg}}^{(n)})^\gamma$, particularly when $p_{\text{neg}}^{(n)}$ is large—that is, when the similarity to a negative-class variation vector is undesirably strong. Conventional focal modulation [36] relies solely on softmax-based probabilities, which limits discrimination in boundary regions where the gap between positive and negative similarities is subtle. To better distinguish such cases, the per-pixel ratio $R^{(n)} = p_{\text{neg}}^{(n)} / p_{\text{pos}}^{(n)}$ is used to select hard negatives. Hard-negative pixels are collected in $\mathcal{H} = \{n \mid R^{(n)} > \tau_R\}$ (with $\tau_R = 0.8$ unless stated otherwise). Focusing the repulsion on the selected hard negatives $\mathcal{H}$ tends to strengthen inter-class separation near ambiguous boundaries and reduce the number of epochs to convergence. The repulsion loss is then defined as:

$$L_r = \frac{1}{N^-} \sum_{n \in \mathcal{H}} -M_r^{(n)} \log(1 - p_{\text{neg}}^{(n)}) \quad (6)$$

where $M_r^{(n)}$ is given in (4).

Finally, the Compositional Similarity Loss L_{cs} combines both terms:

$$L_{cs} = L_a + \lambda_r L_r \quad (7)$$

where $\lambda_r > 0$ balances the contributions of attraction and repulsion. Here, we set $\lambda_r = 1.0$.

By applying class-specific focal modulation to negative samples via the per-pixel ratio $R^{(n)}$, the proposed L_{cs} enables fine-grained modeling of embeddings—crucial for accurate class representation in segmentation tasks with complex boundaries and structures. Therefore, L_{cs} leverages a factorized similarity score combining a representative proxy and variation vectors, while coupling attraction over correct-class variations with focused repulsion on the selected hard negatives $\mathcal{H}$; this preserves intra-class compactness and enforces inter-class separation, yielding more reliable decisions near ambiguous boundaries.

4. Experiments

To validate the versatility and applicability of the proposed method, we conduct experiments on three datasets with distinct characteristics: COCO-Stuff10k [54] for general semantic segmentation, Cityscapes [41] tailored for autonomous driving scenarios, and iSAID [42] for remote sensing tasks. We apply our method across a broad spectrum of segmentation models, including conventional CNN-based backbones (HRNet [12], DeepLabV3 [9]), lightweight architectures optimized for efficiency (MobileNetV3 [37], PIDNet [30]), and transformer-based approaches (SegFormer [13], SegNeXt [43]). This comprehensive evaluation demonstrates the robustness and adaptability of our approach across diverse domains and architectural paradigms.

4.1. Experimental setups

We conduct experiments on the following three datasets:

- Cityscapes [41] is a benchmark for urban scene understanding, consisting of 5000 finely annotated images at a resolution of $2048 \times$

1024. The dataset is split into 2975 training, 500 validation, and 1525 test images, with 19 semantic classes.

- COCO-Stuff10k [54] extends 10,000 images from MS COCO train2014 with pixel-level annotations, covering 80 thing classes, 91 stuff classes, and an ‘unlabeled’ category. We use 9000 images for training and 1000 for validation.
- iSAID [42] is a large-scale aerial image segmentation dataset containing 2806 high-resolution images with approximately 655,451 object instances across 15 classes. We split the original images into 896×896 patches, resulting in 33,978 training and 11,644 validation samples.

We evaluate the proposed method on six segmentation models: MobileNetV3 [37], PIDNet [30], DeepLabV3 [9], HRNet [12], SegFormer [13], and SegNeXt [43]. The bracketed values denote the training settings in order: backbone, input resolution, optimizer, initial learning rate, and number of training iterations. For Cityscapes [41], training was conducted using MobileNetV3 [M-V3-D8, 1024×512 , SGD, 1e-2, 320 k], PIDNet [PIDNet-M, 1024×1024 , SGD, 1e-2, 120 k], HRNet [V2p-W48, 512×1024 , SGD, 1e-2, 160 k], SegNeXt [SegNeXt-B, 512×1024 , AdamW, 1e-4, 160 k], and DeepLabV3 [R101-D8, 512×1024 , SGD, 1e-2, 80 k]. On COCO-Stuff10k [54], the models were trained with SegFormer [MiT-B4, 512×512 , AdamW, 6e-5, 80 k], HRNet [V2p-W48, 512×512 , SGD, 1e-2, 160 k], and DeepLabV3 [R101-D8, 512×512 , SGD, 1e-2, 80 k]. For iSAID [42], training was executed using HRNet [V2p-W48, 896×896 (patch size), SGD, 1e-2, 160 k] and SegNeXt [SegNeXt-B, 896×896 (patch size), AdamW, 1e-4, 160 k] to assess performance in the remote sensing domain. All experiments were carried out using the MMSegmentation [56] framework on Ubuntu 20.04 with Python 3.9 and CUDA 11.8. Model training and evaluation were conducted on two NVIDIA A6000 GPUs, while FPS measurements for real-time methods (MobileNetV3, PIDNet-M) were performed on an NVIDIA RTX 3090. All hyperparameters (learning rates, batch sizes, optimizer-specific coefficients, loss-weight ratios, etc.) were optimized on each dataset’s validation split via grid search combined with cross-validation. Under this protocol, and in combination with the standard cross-entropy objective, the auxiliary scaling is set to $\lambda_{cs} = 1.0$ by default; where the auxiliary-task balance is sensitive, λ_{cs} is adjusted to stabilize optimization.

4.2. Experimental results

Fig. 4 presents segmentation results on the Cityscapes [41] validation set using HRNet-based models. Here, “HRNet + Rep” denotes the model with only the representative proxy, while “HRNet + Rep + Var” corresponds to our proposed method with both representative proxy and variation vectors. In the first row, both “HRNet” and “HRNet + Rep” incorrectly segment some pixels of the “Train” class as “Bus.” In contrast, the proposed method effectively reduces this error by capturing intra-class variation and reinforcing contextual understanding. In the second row, the superiority of our approach is evident even in local regions. The boundary between the cyclist and the bicycle is more clearly separated, and the distinction between sidewalk and road is best preserved, closely matching the ground truth. This demonstrates that our context-aware modeling does not compromise local detail. Lastly, the third row highlights performance in cases of adjacent semantically similar objects (“Bus-Car-Bus”). Our method segments the small “Car” object in the middle more accurately, indicating improved inter-class separation. Overall, these qualitative results clearly show that the proposed method outperforms baselines in both mask consistency and fine-grained boundary recovery.

Tables 1 and 2 present the quantitative results of applying our method to various backbone models on the Cityscapes [41] validation set. We evaluate under two settings: real-time and non-real-time segmentation, analyzing the impact of integrating the proposed representative proxy and variation vectors into existing representative models.

Table 1

Real-time segmentation results on the Cityscapes validation set.

Method	Backbone	Params (M)	FPS	mIoU (%)
BiSeNetV2 [39]	ResNet-18	–	67.6	75.8
STDC2 [40]	STDC2-Seg75	22.2	58.2	77.0
AFFormer [32]	AFFormer-base	3.0	28.4	78.7
DDRNet [38]	DDRNet-23	20.1	51.4	79.5
MobileNetV3 [37]	M-V3-D8	5.6	31.8	69.5
+ Ours			31.8	72.3 $\uparrow$ 2.8
PIDNet [30]	PIDNet-M	34.4	39.8	80.0
+ Ours			39.9	80.6 $\uparrow$ 0.6

Table 2

Non-real-time segmentation results on the Cityscapes validation set.

Method	Backbone	Params (M)	mIoU (%)
Segmenter [44]	ViT-L / 16	322.0	80.7
OCRNet [45]	HRNet-W48	70.5	81.1
SETR [3]	ViT-Large	318.3	82.2
HRFormer [46]	HRFormer-B	56.2	82.6
HRNet [12]	HRNetV2-W48	65.9	80.3
+ Ours			81.4 $\uparrow$ 1.1
SegNeXt [43]	MSCAN-B	48.9	82.5
+ Ours			82.9 $\uparrow$ 0.4

In the real-time setting, our method improves the mIoU of MobileNet [37]—a lightweight backbone with only 6 M parameters—by approximately 2.8 percentage points. This demonstrates that the proposed components are particularly effective in enhancing contextual understanding and modeling intra-class variation in compact architectures. For PIDNet-M [30], which already adopts a strong multi-branch strategy, the performance gain is more modest. However, it still surpasses previous state-of-the-art results without increasing parameter count or computational overhead, which highlights the efficiency of our approach. All real-time evaluations—both with MobileNet and PIDNet-M—are conducted under >30 fps requirement, aligning with established benchmarks in lightweight segmentation research.

In the non-real-time setting, both HRNet [12] and SegNeXt [43] demonstrate consistent performance gains, outperforming prior top-performing methods such as HRFormer [46] and SETR [3]. For HRNet in particular, the improvements in boundary recovery and small-object segmentation illustrated in Fig. 4 are now quantitatively validated.

Table 3

Quantitative results on the COCO-Stuff10k dataset.

Method	Backbone	Params (M)	mIoU (%)
AFFormer [32]	AFFormer-base	3.0	35.2
SVCNet [47]	ResNet-101	–	39.6
DANet [48]	ResNet-101	69.1	39.7
ACNet [55]	ResNet-101	–	40.1
OCRNet [45]	HRNetV2-W48	70.3	40.5
HRNet [12]	HRNetV2-W48	65.9	36.2
+ Ours			37.2 $\uparrow$ 1.0
DeepLabV3 [9]	ResNet-101	68.6	37.7
+ Ours			38.4 $\uparrow$ 0.7
SegFormer [13]	MiT-B4	64.1	46.5
+ Ours			47.1 $\uparrow$ 0.6

Notably, all improvements were achieved without increasing the number of trainable parameters. As seen in Fig. 4, the representative proxy helps refine class-centric embeddings, while the variation vectors enrich local-global interactions, resulting in improved boundary precision and contextual consistency.

In summary, the proposed method offers significant benefits in lightweight and real-time scenarios, yields measurable upper-bound improvements for already strong models, and maintains parameter efficiency throughout. It effectively enhances both local detail and global context without sacrificing model compactness.

Table 3 presents the quantitative comparison results on the COCO-Stuff [54] test set. This benchmark contains 91 stuff classes with extensive background regions and high scene variability, making global contextual understanding more critical than in Cityscapes [41]. For evaluation, we adopt widely used semantic segmentation backbones—HRNet [12], DeepLabV3 [9], and SegFormer [13]—each representing distinct architectural paradigms: high-resolution multi-branch [12], dilated convolution-based encoder-decoder [9], and Transformer-based encoder-decoder [13]. This setup allows us to analyze how the proposed approach interacts with differing backbone designs.

With our training scheme applied, HRNet improves from 36.2–37.2 mIoU, outperforming comparable state-of-the-art methods. DeepLabV3 also shows a gain from 37.7–38.4, indicating that even dilated CNN-based architectures benefit from our method. SegFormer improves from 46.5–47.1, pushing the absolute performance of Transformer-based models further.

Due to its complex background and extreme scene variation, COCO-Stuff typically yields limited absolute gains. Nevertheless, consistent

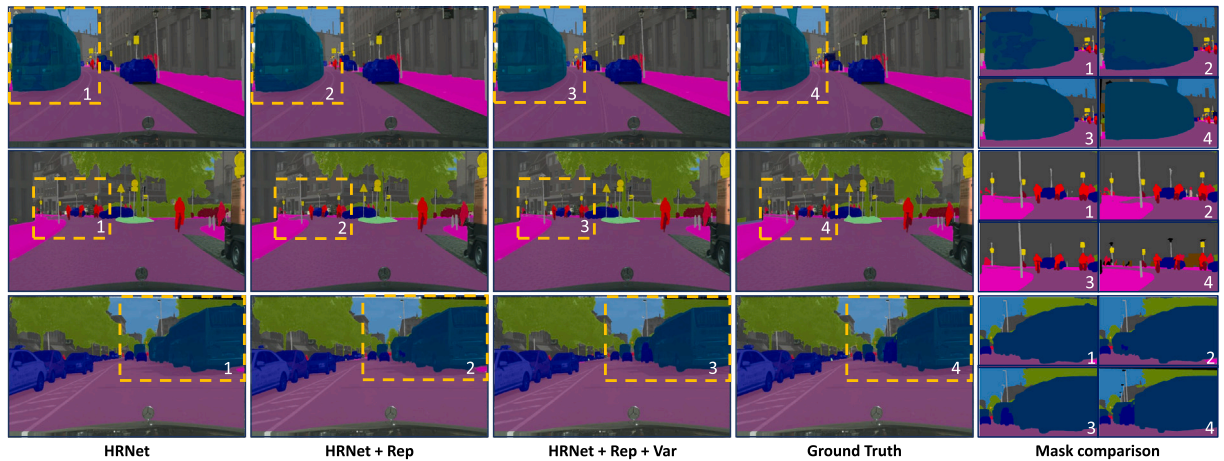


Fig. 4. Qualitative results on the Cityscapes validation set. From left to right: predictions from the HRNet [12] baseline, HRNet with only the representative proxy (HRNet + Rep), HRNet with both representative proxy and variation vectors (HRNet + Rep + Var, our method), and the GT overlaid on the input image. The rightmost column shows a zoomed-in comparison of the predicted masks.

Table 4
Quantitative results on the iSAID dataset, a standard benchmark in remote sensing segmentation.

Method	Backbone	mIoU (%)
DenseASPP [11]	ResNet-50	57.3
PSPNet [2]	ResNet-50	60.3
BSNet [51]	ResNet-50	63.5
FarSeg [53]	ResNet-50	63.7
RSSFormer [50]	RSS-B	65.9
PFNet [52]	ResNet-50	66.9
AerialFormer [49]	Swin-B	69.3
HRNet [12]	HRNetV2-W48	62.3
+ Ours		64.2 $\uparrow$ 1.9
SegNeXt [43]	MSCAN-B	69.9
+ Ours		70.2 $\uparrow$ 0.3

improvements across all three architecturally diverse backbones suggest that the representative proxy effectively refines class-centric embeddings, while the variation vectors complement surrounding context—together enhancing prediction confidence even in challenging backgrounds. These results demonstrate that our method functions not only as an efficient real-time solution but also as a general-purpose loss formulation that improves performance across mainstream CNN and Transformer backbones without increasing parameters.

Lastly, Table 4 shows results from the remote sensing benchmark iSAID [42], evaluating the generalization capability of our method. When applied to two different backbones [12,43] under identical training settings, our approach achieves gains of +1.9 and +0.3 mIoU, respectively. This highlights the effectiveness of our representation enhancement strategy across both natural and aerial imagery, underscoring the potential of the proposed approach as a domain-agnostic solution for semantic segmentation.

4.3. Ablation studies

To comprehensively evaluate the effectiveness of the proposed framework, we conduct a series of ablation studies spanning multiple datasets and backbone architectures. We first analyze the effect of varying the number of class-specific variation vectors in HRNet [12] on Cityscapes [41], examining how this design choice influences segmentation accuracy and model complexity. Building on this, we investigate the joint impact of focal modulation and variation vectors by comparing alternative proxy configurations under strictly matched conditions on ADE20K [60] and iSAID [42] with both HRNet [12] and PIDNet [30] backbones. We then disentangle the contribution of individual loss components—attraction, repulsion, and focal modulation—using a lightweight MobileNetV3 [37] backbone to highlight their relative influence in isolation. Furthermore, we examine the sensitivity of the Compositional Similarity Loss to the hard-negative mining threshold τ_R on both Cityscapes and iSAID, identifying the optimal range that balances stable training with discriminative capability. Finally, we provide t-SNE visualizations of the learned embeddings, which reveal progressively sharper inter-class boundaries as training advances, thereby demonstrating the capacity of our method to enhance feature separability in the latent space.

4.3.1. Effect of varying class-specific variation vectors

Table 5 reports the results of an ablation study on the Cityscapes validation set by incrementally increasing the number of class-specific variation vectors in the HRNet [12] backbone. Expanding the number of variation vectors (K_c) up to 5 leads to consistent performance gains, as the enriched class representations allow for finer prediction boundaries. However, beyond 5 variation vectors, the performance saturates, suggesting diminishing returns due to redundant information among excessive auxiliary embeddings. While the number of variation vectors increases, the model’s parameter count grows only marginally. As shown

Table 5
Ablation analysis on varying the number of variation vectors per class.

K_c	Params (K)	mIoU (%)
0	610	80.7
2	614	81.0
3	616	81.0
5	621	81.4
10	633	81.1

Table 6

Multi-proxy ablation highlighting the effects of focal modulation (FL) and variation vectors under matched training.

Method	Proxy	FL	$\vec{P}_c$	$\vec{v}_{c,i}$	mIoU (%)		
					ADE20K	iSAID	
					HRNet	PIDNet	HRNet
Baseline			0	0	42.0	38.8	62.3
Zhou22 [22]			10	0	43.0	39.5	63.8
Ours (rep-only)	✓	✓	6	0	42.7	39.2	63.4
Ours (rep + var)	✓	✓	1	5	43.2	39.9	64.2

Table 7

Component ablation on Cityscapes validation set.

Method	L_a	L_r	Focal	mIoU (%)
Baseline				69.5
+ Attraction	✓			71.7
+ Repulsion		✓		71.1
+ Attraction / Repulsion	✓	✓		72.0
+ Ours	✓	✓	✓	72.3

in the table, expanding from 2 to 10 vectors results in an increase from 614 K to just 633 K parameters—a mere 3 % rise. This is because the projection head and representative proxies remain fixed, and only the variation vector tensors grow. Importantly, these additional parameters are only active during training and are excluded from the inference graph. Therefore, the enhanced representational capacity does not incur additional cost in model size or computation during testing.

4.3.2. Evaluating multi-proxy strategies: focal modulation and variation vectors

Table 6 compares three multi-proxy add-ons under strictly matched settings. All methods share the same backbone and training pipeline, and we keep the train/val split and augmentation consistent across datasets (ADE20K and iSAID). For Zhou et al. [22], we apply the hyper-parameter settings reported in the paper; when items are unspecified, we maintain consistency without additional tuning. Our methods use the same training recipe without method-specific tuning. Although all compared methods adopt a proxy-centered push–pull learning regime, they instantiate it differently. Zhou’s method combines a standard pixel–proxy contrast with a compactness term and, in our evaluation, uses ten representative proxies per class. Our variants adhere to the same general framework while shaping their dynamics in complementary ways: the rep-only setting couples a negative-focused focal modulation (FL) with six representative proxies, whereas the rep + var setting retains FL and explicitly factorizes the capacity into one representative proxy and five variation vectors. In Table 6, the FL column indicates whether the negative-focused focal modulation is used, while the $\vec{P}_c$ and $\vec{v}_{c,i}$ columns report the per-class counts of representative proxies and variation vectors, respectively. Empirically, proxy-based training yields improvements over the baseline; adding FL confers consistent additional gains; and incorporating variation vectors delivers the largest incremental improvement across the HRNet/PIDNet configurations on

Table 8
Ablation analysis of the hard-example threshold τ_R .

Dataset	Model	τ_R	mIoU (%)
Cityscapes	PIDNet [30]	0.7	80.5
		0.8	80.6
		0.9	80.4
		1.0	80.4
iSAID	HRNet [12]	0.7	64.1
		0.8	64.2
		0.9	64.2
		1.0	64.0



Fig. 5. Visualization of variation vector assignments for the Bus (left) and Rider (right) classes on Cityscapes. Each color indicates the assignment of pixels to a distinct variation vector within the class, illustrating how the proposed factorization captures intra-class diversity. For clarity, only three variation vectors are visualized per class.

both datasets (ADE20K and iSAID). Although ADE20K is the more challenging dataset in our setting, these advantages persist there and also transfer to the distinct aerial-domain iSAID, indicating that—beyond merely increasing the number of proxies—focal weighting and capacity factorization materially influence segmentation quality.

4.3.3. Effect of loss components on the proposed objective

To clarify the individual contribution of each component in the proposed objective—Attraction (L_a), Repulsion (L_r), and focal modulation—we systematically enable or disable them while keeping all other training conditions identical. Experiments are conducted on the Cityscapes validation set with a MobileNetV3 backbone, chosen as a lightweight architecture where the relative impact of each component is most apparent. As summarized in Table 7, the variant with L_a alone leads to reduced intra-class variance, while the variant with L_r alone proves less effective in the absence of a positive anchor. Incorporating both L_a and L_r yields a more substantial gain by jointly promoting compact clusters and enlarging inter-class margins. The full configuration, which further integrates focal modulation to emphasize hard negatives near decision boundaries, achieves the best performance, indicating that the proposed approach benefits from selectively focusing repulsion rather than applying it uniformly.

4.3.4. Impact of hard-negative mining threshold τ_R

Table 8 summarizes the effect of the threshold τ_R —used for selecting negative candidates in the computation of the Compositional Similarity Loss—on validation performance for Cityscapes and iSAID. Across both backbones, the highest mIoU was observed when τ_R was set within the range of 0.8–0.9, with PIDNet [30] reaching up to 80.6. This suggests that excluding overly easy negatives while retaining sufficiently “hard” examples concentrates the loss signal and leads to improved learning. Conversely, when $\tau_R = 1.0$ (i.e., no hard mining), the effect of negative selection diminished, resulting in performance degradation. Setting τ_R too low (e.g., below 0.6) forced excessive separation from negative embeddings, leading to gradient explosion and significant training instability. Therefore, these findings indicate that setting τ_R near 0.8 offers the best trade-off—effectively leveraging hard-example mining while maintaining stable optimization.

4.3.5. Visualization of variation vector assignments

To further examine the contribution of the proposed variation vectors to modeling intra-class diversity, we present qualitative visualizations in Fig. 5 using an HRNet-based model trained on the Cityscapes dataset. The left column illustrates examples from the Bus class, where different vectors are responsible for distinct regions such as the windows or the lower body of the vehicle. The right column shows examples from the Rider class, where the vectors capture localized structures, including the

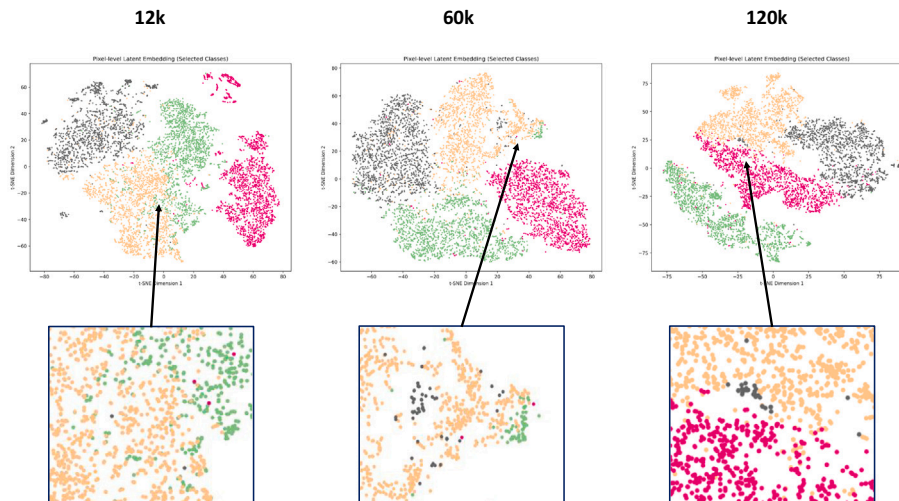


Fig. 6. The results of PIDNet trained with the proposed learning method. We visualize the model’s output for the “Car,” “Truck,” “Bus,” and “Train” classes on the validation set by mapping the embeddings to a latent space at 12 k, 60 k, and 120 k training iterations using t-SNE.

head and lower body. Taken together, these results demonstrate that the variation vectors consistently preserve intra-class diversity by attending to complementary aspects within the same category, thereby offering interpretable evidence of the effectiveness of the proposed factorized representation.

4.3.6. t-SNE visualization of progressive class separation

Fig. 6 visualizes the latent space of PIDNet [30] trained with the proposed method, using t-SNE on validation data. From left to right, the figure shows the distribution of embeddings at 12 k, 60 k, and 120 k training iterations. The zoomed-in regions highlight areas with relatively poor class separation, clearly illustrating how overlapping regions in the early stages become more distinct as training progresses. As training proceeds, the data points gradually form well-separated clusters, indicating that the model is learning more discriminative feature representations over time. Overall, this visualization demonstrates that the proposed learning strategy effectively enhances inter-class boundaries in the latent space, leading to improved class separability.

5. Conclusion

In this paper, we propose a novel representation learning framework called Variation-Aware Proxy Learning for semantic segmentation. By jointly learning a representative proxy to capture the core semantics of each class and variation vectors to model intra-class variations, we introduce a factorized similarity score that enables fine-grained representation learning. To better handle hard examples near class boundaries, we incorporate focal modulation into this score, and design Attraction and Repulsion Losses to enhance intra-class compactness and inter-class separability. The proposed method requires no additional inference-time parameters and can be seamlessly integrated into a wide range of segmentation backbones, including both CNN and Transformer architectures. Extensive experiments on Cityscapes, COCO-Stuff10k, iSAID, and ADE20K datasets demonstrate consistent mIoU improvements, particularly in regions with high intra-class diversity or ambiguous class boundaries. Moreover, the method proves to be model-agnostic, offering benefits across both lightweight real-time models and high-capacity networks. Future work will explore automated tuning of hyperparameters such as the number of variation vectors and the hard-example mining threshold, as well as lightweight variation vector designs to further improve training stability and efficiency.

CRediT authorship contribution statement

Haejun Bae: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis.
Byung Cheol Song: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered potential competing interests:

Byung Cheol Song reports that financial support was provided by the Korean government (MSIT).

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Data availability

The authors do not have permission to share data.

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